To: Editor-in-Chief,
Computerized Medical Imaging and Graphics (Elsevier)

Date: October 27, 2026
Subject: Submission of Original Research Article

Dear Editor-in-Chief and Editorial Board,

We are pleased to submit our original research manuscript entitled:

"Calibration-Aware Anisotropic 3D Hysteresis and Leakage Auditing for Volumetric Lower-Grade Glioma Segmentation and Radiogenomic Profiling in Brain MRI"

authored by Shayan Ganji Nataj and Keivan Borna (Corresponding Author), for publication consideration as an original research article in *Computerized Medical Imaging and Graphics*.

Lower-grade gliomas (LGG; WHO grades II and III) represent diffusely infiltrative primary brain neoplasms where accurate computerized delineation from multi-sequence MRI is indispensable for maximal neurosurgical resection, conformal radiotherapy, and non-invasive molecular characterization. While 2D deep learning segmentation models frequently report high overlap scores on benchmark datasets, their translational deployment is severely hindered by two unaddressed methodological vulnerabilities: (1) slice-level data leakage from naive random partitioning, where networks memorize subject-specific features across contiguous slices, and (2) severe probability miscalibration induced by overlap loss formulations (e.g., Focal Tversky), resulting in overconfident probability halos and excessive false-positive alarms on healthy, non-tumor slices (which comprise 65.05% of the TCGA-LGG imaging volume).

To bridge methodological innovation with clinical translation—a core mission of *Computerized Medical Imaging and Graphics*—we introduce **CAL-3DH** (*Calibration-Aware, Leakage-Audited Anisotropic 3D Hysteresis*), an end-to-end volumetric framework combining quality-aware preprocessing, nested probability calibration, and anisotropic 3D spatial post-processing. Evaluated across the complete TCGA-LGG cohort (110 patients, 3,929 axial MRI slices) under patient-grouped 5-fold cross-validation with nested inner-fold tuning, the major contributions of this study are:

1. **Rigorous Leakage Quantification:** Through a controlled 5-fold audit under matched hyperparameters, we prove that conventional random slice splitting artificially inflates positive-slice Dice by **+8.79%** (73.96% vs. 82.74%) and overall Dice by **+6.20%** via patient identity memorization, explaining widespread failure of published 2D models on external clinical cohorts.

2. **Nested Uncertainty Calibration:** Integrating scalar temperature scaling ($T^* = 1.844$) fitted strictly on internal validation logits (`inner_val`), we reduce pooled Expected Calibration Error (ECE) by **46.9%** (0.369% to 0.196%) and contract the active margin overconfidence gap by 20.9% on unseen test data without altering classification boundaries.

3. **Anisotropic 3D Z-Continuity Hysteresis:** By decoupling planar contouring ($\tau^*_{xy} = 0.74$) from axial boundary rescue along the z-axis ($\tau^*_{z,\text{low}} = 0.50$) with spatial volume filtering ($V^*_{\min} = 38$), CAL-3DH suppresses empty-slice false alarms by **42.4%** relative to baseline (from 349 to 201 slices) while recovering missed apical slices, achieving $\text{Dice}_{\text{pos}} = \mathbf{74.92}\%$ [95% CI: 70.82%–78.39%] and $\text{Dice}_{\text{all}} = \mathbf{86.12}\%$ [95% CI: 84.32%–87.75%].

4. **Patient-Level 3D Agreement and Radiogenomic Fidelity:** Patient-level volumetric analysis ($n = 110$) demonstrates high clinical agreement with radiologist ground truth (two-way random absolute agreement $\text{ICC}(2,1) = \mathbf{0.8894}$, Pearson $r = 0.8886$, Bland-Altman mean

volumetric bias = +143.7 voxels, HD_{95} = 10.97 voxels). Crucially, downstream radiogenomic profiling confirms that CAL-3DH-segmented 3D volumes preserve statistically significant stratification across TCGA molecular consensus clusters (*COCCluster*, Kruskal-Wallis $p = \mathbf{0.0268}$) with exact biological rank ordering preservation (COC 1 > COC 3 > COC 2).

We declare that:

- This manuscript represents original work that has not been published previously and is not currently under consideration for publication elsewhere.
- Publication is approved by all authors and by the responsible institutional authorities where the research was performed.
- The authors declare no competing financial or personal interests that could influence the work reported herein.
- To promote full scientific reproducibility and open science in computational neuro-oncology, the complete codebase, patient-grouped 5-fold cross-validation splits, precomputed features, and auditing scripts are openly available on GitHub at: https://github.com/Shayan-Ganji/lgg_segmentation.

Thank you very much for your time and for considering our work for publication in *Computerized Medical Imaging and Graphics*.

Sincerely,

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